



The Shed, explained

A file and folder browser on the starter you sent, built the way I build my own project: shape and tests first, every decision written down, and a live site.

MAPLARGE DEVELOPER TEST PROJECT | STEVEN STOUT | 29 SEPTEMBER 2026

LIVE theshed.stevenstout.biz

CODE github.com/SteveStout/TheYard, [samples/maplarge](#)

FULL SIZE theyard.stevenstout.biz

RESUME AND ABOUT on the site, one click each

What it is

One configurable home directory, one API, one page. The address bar is the whole state.

An ASP.NET Core 10 API over a home directory set by configuration, and a page in TypeScript with no framework. It browses, searches by name or wildcard, uploads with a progress bar, downloads, makes folders, moves, copies and deletes, inside a native dialog with one trigger. Every failure is a sentence the page shows. Fourteen decision records are served from the running app, each with the code it decided about read from the build as you read it.

QUICK START

```
dotnet run          # localhost:5120, browsing sample-home/
dotnet test         # 130 xUnit tests, warnings as errors
npm install && npm test # 21 node tests over the compiled modules
```

OR SKIP THE BUILD, THREE LINKS

- theshed.stevenstout.biz/?path=reports/2026: the dialog, open in a folder.
- [/?q=*.csv&sort=size&dir=desc](https://theshed.stevenstout.biz/?q=*.csv&sort=size&dir=desc): a search, sorted, as a link.
- [/?view=docs&doc=adr-003-the-line-a-path-cannot-cross](https://theshed.stevenstout.biz/?view=docs&doc=adr-003-the-line-a-path-cannot-cross): the security record, its code live. Read ADR-003, 005 and 009 first; the README is the map.

The brief, and where each item is

You asked for	Where it is, under samples/maplarge (its code: pages 2 to 4)
A. Builds in VS, Rider, VS Code, CLI	TestProject.sln, your file kept by name, on .NET 10; no package.
B. JSON API: browse and search	Controllers/FilesController.cs; every failure an RFC 9457 document from Controllers/ProblemResponseHandler.cs.
C. Easy folder configuration	Files:Home in appsettings.json, or FILES__HOME on a server; blank shows sample-home. Nobody gets outside it.
D. Deep-linkable state in the URL	src/lib/urlState.ts parses and writes six keys; src/navigation.ts has the one navigate().
E. Pure TypeScript	src/ compiled by tsc alone to wwwroot/js; no framework, no bundler.
F. Upload and download	src/lib/api.ts uploads with a bar per file; downloads stream with ranges.
G. Counts and sizes per view	Domain/ViewTotals.cs, on every reply from Application/FileBrowser.cs, plus took_ms.
H. Bonus: dialog, writes, performance	A native dialog in wwwroot/index.html; writes ask in the row; tests/TestProject.Tests/PerformanceTests.cs.

How it was built and shipped

- **Green before every commit.** 151 tests, warnings as errors, and a headless Chrome pass of the running page (deep links, search, Back, upload, delete, Escape) with zero console errors.
- **Live at no added cost.** A third web app on the App Service plan my project runs on, its own free managed certificate, the image in the same registry, rolled by a workflow whose identity holds Website Contributor on that one site and nothing wider. The Bicep and the workflow are in the folder.
- **Built with AI, and written down (ADR-011).** I decided the shape and the test list first, Claude drafted against them, the build ran on my machine, and I read everything before it shipped. My first push had a folder-name case collision Windows hides and Linux does not; the container found it, and the link test now checks case exactly.

By the numbers

151 tests, green before every commit	< 1 ms to browse 100 files, measured by a test	44 ms to search 10,000 files whole; capped, 8 ms
0 warnings, analyzers at their strictest	14 records served live with their code	\$0 added hosting, on a plan I already pay for

Over 10,000 files on the build machine (ADR-008)

lower is better

Browse, 100 files		< 1 ms
Search, capped at 200		8 ms
Search, whole tree		44 ms

Six choices, and why

1 Your starter, kept Your names, your controller shape, your one page. Three lines added to the project file (ADR-001).	2 Dependencies inward Onion architecture in one project, four folders. A test fails any dependency that points outward (ADR-013).
3 One pure security class The rule sits in the innermost layer: three string refusals, then containment. Tested with no disk (ADR-003).	4 The address is the state Nothing is kept anywhere else. The first keystroke of a search pushes; the rest replace (ADR-005).
5 Measured, not claimed Search is lazy, stops at a cap and says so; a test prints the timings on every run (ADR-008).	6 Rules beside their tests A table of every standing rule and the test that holds it; a test reads the table (ADR-009).

Taking it further: my larger project is the worked example

TheYard ([start here](#)) is the same practice at full size: .NET 10 and React, Azure SQL and Cosmos DB, 90 records, about 2,100 test runs per gate, two sites on one plan. Each step below is something the Yard already does.

Next on the Shed	How the Yard already does it
A committed browser test	143 Playwright specs in the gate; axe holds eleven views to WCAG 2.1 AA. TheYard: the accessibility check
Paging behind the search cap	The API pages 100,000 vehicles behind Load more; the page never asks for them all. TheYard: the vehicles API
Logs that outlive the container	Every request and error kept in Cosmos DB for three years (ADR-071, 073). TheYard: the kept logs
The path guard as a Files card	The Admin tab is a product: health, machines, timing, logs, each a card (ADR-080). TheYard: the Admin tab

A. Builds in VS, Rider, VS Code, CLI

It opens and builds in Visual Studio, Rider, VS Code or the command line on .NET 10, with no extra packages, and your solution and project file keep their names. Three settings were added: warnings stop the build, the code checkers are on, the documents ship with the app. Why it matters: any developer opens it in their own tool, and problems surface on their machine before a reviewer sees them. [TheYard: The rules a change has to pass](#)

samples/maplarge/TestProject.csproj

```
<!-- The starter's project file, kept as MapLarge sent it (nullable, implicit usings) on net10.0, the
     same framework as TheYard (ADR-001), with three additions: warnings are errors, the sealed-by-default analyzer is on, and the
     documents plus the sample home travel with a publish so the container serves them (ADR-001). -->
<PropertyGroup>
  <TargetFramework>net10.0</TargetFramework>
  <Nullable>enable</Nullable>
  <ImplicitUsings>enable</ImplicitUsings>
  <TreatWarningsAsErrors>true</TreatWarningsAsErrors>
  <EnforceCodeStyleInBuild>true</EnforceCodeStyleInBuild>
  <AnalysisLevel>latest-recommended</AnalysisLevel>
  <GenerateDocumentationFile>true</GenerateDocumentationFile>
  <InvariantGlobalization>true</InvariantGlobalization>
</PropertyGroup>
```

B. JSON API: browse and search

The page talks to the server through a small JSON API, one line per route in Controllers/FilesController.cs. When something goes wrong, the answer always has the same shape: a standard error document (RFC 9457) from Controllers/ProblemResponseHandler.cs (piece 6). Why it matters: a developer can call it from any tool without reading the code, and every team that uses it handles errors one way. [TheYard: The API describes itself](#)

samples/maplarge/Controllers/FilesController.cs, Browse and Search

```
/// <summary>Lists one folder's contents. GET /api/files?path=docs</summary>
/// <param name="path">A folder, relative to home; empty means home.</param>
[HttpGet]
public Listing Browse([FromQuery] string? path) => browser.Browse(path);

/// <summary>
/// Finds everything under a folder whose name matches the query. GET /api/files/search?path=&q=*.md&limit=200
/// </summary>
/// <param name="path">The folder to search under; empty means home.</param>
/// <param name="q">A substring, or a glob pattern using * and ?.</param>
/// <param name="limit">The maximum number of matches to return.</param>
[HttpGet("search")]
public SearchResult Search([FromQuery] string? path, [FromQuery] string? q, [FromQuery] int? limit) => browser.Search(path, q, limit);
```

C. Easy folder configuration

The app shows one folder. Pick it in appsettings.json on the line Files:Home, or on a server with the setting FILES__HOME, which wins. Pick nothing and it shows the practice folder that ships with it, sample-home. Whichever folder it is, nobody gets outside it: Domain/HomePath.cs (piece 4) is the guard. Why it matters: a developer clones and runs with no setup, and an organization ships one build to every server, each pointed at its own folder. [TheYard: The code is public and the secrets are not](#)

samples/maplarge/appsettings.json and Composition/FilesRegistration.cs, HomeFor

```
"Files": {
  "Home": "",
  "MaxUploadBytes": 104857600,
  "SearchLimit": 200,
  "SearchCeiling": 1000
}

/// <summary>The practice folder that ships with the project, shown when no folder is set.</summary>
private const string PracticeFolder = "sample-home";

/// <summary>
/// Picks the folder the app shows and hands it to HomePath, the guard that keeps every request inside it. The folder comes from Files:Home in
/// appsettings.json, or from the FILES__HOME environment variable, which wins. Nothing set means the sample-home practice folder.
/// </summary>
/// <param name="options">The Files section of configuration.</param>
/// <param name="contentRoot">The folder the project runs from.</param>
public static HomePath HomeFor(FilesOptions options, string contentRoot)
{
  // Which folder was chosen? Nothing chosen means the practice folder.
  string chosen = string.IsNullOrEmpty(options.Home) ? PracticeFolder : options.Home;

  // A full path (C:\files or /srv/files) is used as written. A short one (my-files) is taken from the folder the project runs from.
  string folder = Path.IsPathRooted(chosen) ? chosen : Path.Combine(contentRoot, chosen);

  // Make the folder if it is not there yet, so a fresh copy runs with no setup.
  Directory.CreateDirectory(folder);

  // From here on, HomePath is the only code that turns a request into a real path on disk, and it refuses any path that would leave this folder.
  return new HomePath(folder);
}
```

D. Deep-linkable state in the URL

Everything on screen (the folder, the search, the sort) is written in the address bar. `src/lib/urlState.ts` reads and writes it (piece 8), and `src/navigation.ts` changes it in one place (piece 9). A setting left at its default is not written, so links stay short. Why it matters: copy the address and anyone opens exactly what you see, so a bug report carries its own steps, and Back and Forward just work. [TheYard: Every diagram opens on its own page](#)

`samples/maplarge/src/lib/urlState.ts`, `DEFAULTS`

```
/**
 * The value of each key when the address does not set it: the file browser view, the top folder, no search, no document, sorted by name in ascending
 * order. An address with none of these keys means the dialog is closed.
 */
export const DEFAULTS: Readonly<Omit<State, 'open'>> = Object.freeze({
  view: 'browse',
  path: '',
  q: '',
  sort: 'name',
  dir: 'asc',
  doc: '',
});
```

E. Pure TypeScript

Plain TypeScript, compiled by `tsc` alone: no framework, no bundler. `src/lib` holds the logic and `src/ui` draws the screen. Why it matters: a new developer reads it top to bottom, and there is no framework upgrade to schedule. [TheYard: The React configuration, explained](#)

`samples/maplarge/tsconfig.json` and `wwwroot/index.html`

```
// TypeScript with nothing else (ADR-006): the sources in src/ compile to plain ES modules in wwwroot/js, which the page loads as they are. No
// bundler, no framework. The output is committed so `dotnet run` works with the .NET SDK alone; `npm run build` regenerates it.
"compilerOptions": {
  "target": "ES2022",
  "lib": ["ES2022", "DOM", "DOM.Iterable"],
  "module": "ES2022",
  "moduleResolution": "bundler",
  "rootDir": "src",
  "outDir": "wwwroot/js",
  ...
},
"include": ["src/**/*.ts"]
}

<script type="module" src="/js/main.js"></script>
```

F. Upload and download

Every upload shows its own progress bar (`src/lib/api.ts` uses `XMLHttpRequest`, because `fetch` cannot report upload progress). Downloads stream, and a broken one can pick up where it stopped. Why it matters: a user sees a big upload moving instead of a frozen screen, and a dropped download on a slow line resumes instead of starting over. [TheYard: Cache headers](#)

`samples/maplarge/src/lib/api.ts`, `upload` and `Controllers/FilesController.cs`, `Download`

```
/**
 * Uploads one file and reports progress while it goes. It uses XMLHttpRequest because fetch cannot report upload progress, and a large file needs a
 * progress bar so the person can see it is still moving.
 * @param path the folder to upload into
 * @param file the file to send
 * @param overwrite true to replace a file with the same name; with false the server answers 409
 * @param onProgress called with the fraction sent so far, from 0 to 1
 */
export function upload(path: string, file: File, overwrite: boolean, onProgress: (fraction: number) => void): Promise<TransferResult> {
  return new Promise((resolve, reject) => {
    const form = new FormData();
    form.append('files', file, file.name);
    const request = new XMLHttpRequest();
    request.open('POST', `/api/files/upload?path=${encodeURIComponent(path)}&overwrite=${overwrite ? 'true' : 'false'}`);
    request.upload.addEventListener('progress', (event) => {
      if (event.lengthComputable) {
        onProgress(event.loaded / event.total);
      }
    });
  });
  ...
}

/// <summary>
/// Sends a file's bytes as a download. Range requests are enabled and the file's last change time is sent with it, which is what a browser needs to
/// resume a paused or broken download where it stopped. The stream comes from the file store and is closed once the response is sent.
/// </summary>
/// <param name="path">A file, relative to home.</param>
[HttpGet("download")]
public IActionResult Download([FromQuery] string? path)
{
  (Stream content, FileEntry entry) = browser.Download(path);
  return new FileStreamResult(content, "application/octet-stream")
  {
    FileDownloadName = entry.Name,
    LastModified = DateTimeOffset.FromUnixTimeMilliseconds(entry.ModifiedMs),
    EnableRangeProcessing = true,
  };
}
```

G. Counts and sizes per view

Every listing says how many folders and files it shows and how big they are, added up by Domain/ViewTotals.cs from exactly what is listed, plus took_ms, how long the server took. Why it matters: a user sees the size of a folder at a glance, and a developer sees the server's own timing on every reply, so a slow request is spotted without extra tools. [TheYard: The listing that went stale while you looked at it](#)

samples/maplarge/Domain/ViewTotals.cs and Application/FileBrowser.cs, Browse

```
/// <summary>Counts the folders and files and adds up the files' sizes.</summary>
/// <param name="folders">The folders in the result.</param>
/// <param name="files">The files in the result.</param>
public static Totals Of(IReadOnlyList<FolderEntry> folders, IReadOnlyList<FileEntry> files)
{
    long bytes = 0;
    foreach (FileEntry file in files)
    {
        bytes += file.SizeBytes;
    }
    return new Totals(folders.Count, files.Count, bytes);
}

/// <summary>
/// Lists one folder's direct contents with totals and timing. Folders come first, then files, and each group is sorted by name ignoring case.
/// </summary>
/// <param name="path">The folder, relative to home; "" means home itself.</param>
public Listing Browse(string? path)
{
    long started = Stopwatch.GetTimestamp();
    string absolute = Folder(path);
    var folders = new List<FolderEntry>();
    var files = new List<FileEntry>();
    foreach (StoreEntry entry in store.Children(absolute))
    {
        Sort(entry, folders, files);
    }
    folders.Sort((a, b) => ViewTotals.NameOrder.Compare(a.Name, b.Name));
    files.Sort((a, b) => ViewTotals.NameOrder.Compare(a.Name, b.Name));
    string relative = home.Relative(absolute);
    return new Listing(relative, HomePath.ParentOf(relative), folders, files, ViewTotals.Of(folders, files), Elapsed(started));
}
```

H. Bonus: dialog, writes, performance

The browser opens in the browser's own dialog box (wwwroot/index.html), so the keyboard and Escape work with no extra code. Delete, move and copy ask for a yes in the row itself, never a pop-up. tests/TestProject.Tests/PerformanceTests.cs times 10,000 files on a real disk. Why it matters: it is usable by keyboard from day one, a confirmation sits next to the thing it changes, and the speed is measured on every build rather than promised. [TheYard: Same performance, proven](#)

samples/maplarge/wwwroot/index.html and tests/TestProject.Tests/PerformanceTests.cs

```
<dialog class="shed" aria-labelledby="shed-title">
  <div class="panel">
    <div class="dialog-head">
      <h2 id="shed-title">The Shed</h2>
      <div class="seg" id="tabs" role="group" aria-label="View">
        <button type="button" data-view="browse" aria-pressed="true">Files</button>
        <button type="button" data-view="docs" aria-pressed="false">Docs</button>
      </div>
      <span class="spacer"></span>
      <button type="button" id="close-shed" class="close" aria-label="Close the file browser">&times;</button>
    </div>
    <section id="browser" aria-label="Files"></section>
    <section id="docs" class="docs" aria-label="Documents" hidden></section>
  </div>
</dialog>

// Browse once before timing to warm the OS directory cache, so the timings match a repeat visit rather than the first read of a cold disk.
browser.Browse("folder-000");

long browseMs = Time(() => browser.Browse("folder-042"), out int browsed);
long cappedMs = Time(() => browser.Search("", "*.txt", 200), out int capped);
long fullMs = Time(() => browser.Search("", "file-099-099", null), out int found);
long rootMs = Time(() => browser.Browse(""), out int folders);

output.WriteLine($"generated {Folders * FilesPerFolder:N0} files in {generated.ElapsedMilliseconds} ms");
output.WriteLine($"browse one folder of {browsed} files: {browseMs} ms");
output.WriteLine($"browse the root of {folders} folders: {rootMs} ms");
output.WriteLine($"search capped at {capped} matches: {cappedMs} ms");
output.WriteLine($"search the whole tree for one name ({found} match): {fullMs} ms");

Assert.Equal(FilesPerFolder, browsed);
Assert.Equal(200, capped);
Assert.Equal(1, found);
Assert.True(browseMs < 250, $"browsing one folder took {browseMs} ms");
Assert.True(cappedMs < 500, $"a capped search took {cappedMs} ms");
Assert.True(fullMs < 3000, $"a full-tree search took {fullMs} ms");
```

1. Program.cs is a table of contents

Program.cs reads like a table of contents: each line names one part of the app, and the file beside it holds the details. There are no rules in it, only the order things start in. src/main.ts does the same for the page. Why it matters: a new developer learns what the app is made of in one screen, and a reviewer knows exactly where to look for each part. Docs tab record: One project, four folders, dependencies inward. [TheYard: Program.cs, explained for a new developer](#)

samples/maplarge/Program.cs

```

/// <summary>
/// The entry point. It builds the web host and starts it. Each step is a single call, and the comment beside the call names the file that holds the
/// details. The setup lives in those files so that this class stays a short, readable list of what the app is made of.
/// </summary>
public sealed class Program
{
    /// <summary>Builds the web app from its parts, in order, and runs it until shutdown.</summary>
    /// <param name="args">Command line arguments, passed to the host builder.</param>
    public static void Main(string[] args)
    {
        var builder = WebApplication.CreateBuilder(args);

        builder.AddTheShedFiles(); // Composition/FilesRegistration.cs
        builder.AddTheShedDocumentationAndVersion(); // Composition/DocumentationAndVersionRegistration.cs
        builder.AddTheShedApi(); // Composition/ApiRegistration.cs

        var app = builder.Build();
        app.UseTheShedRequestPipeline(); // Composition/RequestPipeline.cs
        app.MapControllers(); // Controllers/: Files, DocumentationAndVersion, Health

        app.Run();
    }
}

```

2. The file browser, registered

This file sets up the file browser: the folder to show (item C), the rules and the upload limit, so an oversized file gets a message that names it. The rules reach the disk only through the IFileStore interface; the sample keeps files on the server's own disk. Why it matters: moving to Azure Blob Storage is one new class and one line here, and nothing else changes. Docs tab records: One project, four folders, dependencies inward; The line a path cannot cross. [TheYard: The composition root, split by job](#)

samples/maplarge/Composition/FilesRegistration.cs, AddTheShedFiles

```

/// <summary>
/// Registers the file browser's services: the home directory and its limits, the file store interface with the class that reads and writes real
/// files, and the FileBrowser that holds the rules. It also sets the server's upload size limits to match the configured maximum.
/// </summary>
public static class FilesRegistration
{
    /// <summary>
    /// Adds the home directory, the file store, the FileBrowser and the upload size limits to the host.
    /// </summary>
    /// <param name="builder">The host being built.</param>
    public static void AddTheShedFiles(this WebApplicationBuilder builder)
    {
        // Reads the "Files" section of appsettings.json into FilesOptions. An environment variable named FILES_ plus the setting (FILES_HOME, for
        // example) replaces the value from the file, which is how a server chooses its folder without editing the code. The options and the HomePath
        // built from them are registered once, so every class sees the same folder.
        builder.Services.Configure<FilesOptions>(builder.Configuration.GetSection(FilesOptions.Section));
        builder.Services.AddSingleton(provider => provider.GetRequiredService<IOptions<FilesOptions>>().Value);
        builder.Services.AddSingleton(provider => HomeFor(provider.GetRequiredService<FilesOptions>(), builder.Environment.ContentRootPath));

        // IFileStore is the interface the rules use to touch the disk, and PhysicalFileStore is the class that reads and writes real files.
        // FileBrowser holds the rules and depends only on the interface, so tests can swap in a different store.
        //
        // This sample keeps files on the server's own disk to stay simple. For production, store them in Azure Blob Storage or another cloud file
        // storage service, the way TheYard keeps its data in managed Azure services rather than on a server. A container's disk is wiped when the app
        // restarts or redeploys and is not shared between instances, so uploads here do not last. The move is one new class that implements
        // IFileStore and this one line.
        builder.Services.AddSingleton<IFileStore, PhysicalFileStore>();
        builder.Services.AddSingleton<FileBrowser>();

        // FileBrowser.UploadAsync checks the exact per-file limit and names the file in its error. The web server and the form parser get that limit
        // plus one megabyte, to allow for the multipart form's own headers and boundaries. A slightly large request therefore still reaches the
        // per-file check and gets a clear message, while a far too large request is cut off by the server before it is read.
        long maxUpload = builder.Configuration.GetSection(FilesOptions.Section).GetValue("MaxUploadBytes", new FilesOptions().MaxUploadBytes);
        long slack = 1024 * 1024;
        builder.WebHost.ConfigureKestrel(kestrel => kestrel.Limits.MaxRequestBodySize = maxUpload + slack);
        builder.Services.Configure<FormOptions>(form => form.MultipartBodyLengthLimit = maxUpload + slack);
    }
    ...
}

```

3. The documents and the version, registered

This file sets up the version number and the documents. The version is the top line of docs/CHANGELOG.md, the commit comes from the deploy, and the documents come from the docs folder. Why it matters: the version on screen can never disagree with the changelog, so support and developers always know exactly which build they are looking at. Docs tab record: Documents served by the app, with live code. [TheYard: Live code samples](#)

samples/maplarge/Composition/DocumentationAndVersionRegistration.cs

```
/// <summary>
/// Registers the two services behind the Docs tab and the version in the page footer. This file does not set the version or write any documentation.
/// It only makes the readers available to the controllers:
/// <list type="bullet">
/// <item>The version number is set by the newest entry at the top of docs/CHANGELOG.md.
/// Adding a line there is how a new version is set.</item>
/// <item>The commit is set by the deploy, which puts it in the SHED_COMMIT environment
/// variable of the container. A local run reads it from the .git folder instead.</item>
/// <item>The documentation is the markdown files in the docs folder (the decision records
/// and the guides) plus README.md. DocumentationAndVersionController serves them to the Docs tab.</item>
/// </list>
/// </summary>
public static class DocumentationAndVersionRegistration
{
    /// <summary>
    /// Adds DocumentationCatalog, which lists the documents in sidebar order and finds one by name, reading them from disk on every request so an
    /// edited file shows on the next page load. Adds VersionInfo, which VersionReader reads once at startup from the changelog and the commit sources
    /// above. Both are singletons rooted at the content root.
    /// </summary>
    /// <param name="builder">The host being built.</param>
    public static void AddTheShedDocumentationAndVersion(this WebApplicationBuilder builder)
    {
        builder.Services.AddSingleton(new DocumentationCatalog(builder.Environment.ContentRootPath));
        builder.Services.AddSingleton(VersionReader.Read(builder.Environment.ContentRootPath));
    }
}
```

4. The line a path cannot cross

This is the guard. Every path a request asks for passes through here, and anything that tries to step outside the chosen folder is refused before the disk is touched. It sits in the innermost layer (onion architecture) with no disk code, so it is tested on its own. Why it matters: the most important security rule in the app is one small class a reviewer can read in a minute. Docs tab record: The line a path cannot cross. [TheYard: Security](#)

samples/maplarge/Domain/HomePath.cs, Resolve

```
/// <summary>
/// Turns a path from a request into an absolute path under the root, or throws <see cref="PathRefusedException"/>. It first refuses three things by
/// looking at the string alone: a rooted path (a drive letter or a leading slash), a "." or ".." segment anywhere (a segment is one name between
/// slashes), and a segment containing a character the operating system does not allow in a name. It then joins the segments onto the root, lets the
/// operating system work out the final full path, and checks that path still lies inside the root. That final check is the safety net for any trick
/// the first three checks do not anticipate. It is business logic in the innermost layer of the onion architecture, so it depends on nothing and its
/// tests run with no disk.
/// </summary>
/// <param name="relative">The path the request sent; null or "" means home itself.</param>
public string Resolve(string? relative)
{
    string trimmed = (relative ?? string.Empty).Trim();
    if (trimmed.Length == 0)
    {
        return Root;
    }
    if (System.IO.Path.IsPathRooted(trimmed) || trimmed.StartsWith('/') || trimmed.StartsWith('\\'))
    {
        throw new PathRefusedException("A path must be relative to the home directory.");
    }
    string[] segments = trimmed.Split(Separators, StringSplitOptions.RemoveEmptyEntries);
    foreach (string segment in segments)
    {
        if (segment is "." or "..")
        {
            throw new PathRefusedException("A path cannot contain '.' or '..' segments.");
        }
        if (segment.IndexOfAny(System.IO.Path.GetInvalidFileNameChars()) >= 0)
        {
            throw new PathRefusedException($"'{segment}' is not a valid name.");
        }
    }
    // Windows drops trailing dots and spaces from a name, so "... " or "x " can come back as the root with a separator on the end. Trimming the
    // separators hands back the root in the one spelling every caller compares against, so home is always recognised as home.
    string joined = System.IO.Path.GetFullPath(System.IO.Path.Combine(Root, System.IO.Path.Combine(segments))).TrimEnd(Separators);
    if (!IsInside(joined))
    {
        throw new PathRefusedException("A path must stay inside the home directory.");
    }
    return IsRoot(joined) ? Root : joined;
}
```

5. The tests that hold it

Each way a path could sneak out has its own test, including the one most guards miss: /home/files-old starts with the same letters as /home/files but is not inside it. Why it matters: nobody can weaken the guard by accident, because the build fails first. [TheYard: The tests, explained for a new developer](#)

samples/maplarge/tests/TestProject.Tests/HomePathTests.cs

```
/// <summary>
/// Checks HomePath, the guard that keeps every requested path inside the home folder. The tests pass plain strings and never touch a disk, with a
/// root in the form of whichever OS runs them. This guard is what stops a request from reading or writing outside the home, so each way of escaping
/// it (dot segments, rooted paths, bad characters, shared prefixes) has its own case. (more in docs/ADR-003-the-line-a-path-cannot-cross.md)
/// </summary>
public sealed class HomePathTests
{
    ...
    [InlineData("../secrets")]
    [InlineData("docs/../../secrets")]
    [InlineData("docs/./notes")]
    [InlineData("")]
    public void A_dot_or_dot_dot_segment_is_refused(string path)
    {
        var refused = Assert.Throws<PathRefusedException>(() => _home.Resolve(path));
        Assert.Contains("'.' or '..'", refused.Message, StringComparison.Ordinal);
    }
    ...
    public void The_root_with_a_prefix_in_common_is_outside()
    {
        // /home/files-old starts with the same characters as /home/files but is a different folder. A plain string prefix check would wrongly call it
        // inside.
        Assert.False(_home.IsInside(_home.Root + "-old"));
        Assert.False(_home.IsInside(Path.Combine(_home.Root + "-old", "x")));
    }
}
```

6. Every failure is a sentence

When a request is refused, this one handler turns it into the right status code and a standard error document (RFC 9457) with a trace id. The controllers have no try/catch. Why it matters: every error reads the same way for the user and the developer, and the trace id points a developer to that exact request. Docs tab record: The wire. [TheYard: Error handling, one shape everywhere](#)

samples/maplarge/Controllers/ProblemResponseHandler.cs

```
/// <summary>
/// Turns a refused request into a clear error reply in the RFC 9457 format, the web standard for errors: a status code, a short title, the reason in
/// one sentence, and a trace id. This is the one place errors get their shape, so no controller needs a try/catch.
/// </summary>
/// <remarks>
/// What each kind of failure becomes, and why:
/// - The app's own refusals (ApiRefusalException) carry their status and title with them.
/// - A path that tries to leave the home folder (PathRefusedException) is always a 400.
/// - A file the disk will not let us touch, because it is read-only or another program has it open, is a 403 or a 409 with the operating system's own
/// message. The request asked for something the disk cannot do; the server is not broken.
/// - Anything else is a real bug. It is left alone here, so it still reaches the default handler as a 500 and is never passed off as the caller's
/// mistake.
/// </remarks>
public sealed class ProblemResponseHandler(IProblemDetailsService problems) : IExceptionHandler
{
    /// <summary>
    /// Called by ASP.NET Core when a request throws. Returns true when it wrote the error reply, or false to pass the exception on to the next
    /// handler.
    /// </summary>
    public async ValueTask<bool> TryHandleAsync(HttpContext httpContext, Exception exception, CancellationToken cancellationToken)
    {
        // 1. Pick the status code and title from the kind of exception. A status of 0 means this is not a refusal this handler knows.
        (int status, string title) = exception switch
        {
            ApiRefusalException problem => (problem.Status, problem.Title),
            PathRefusedException => (StatusCodes.Status400BadRequest, "The path was refused"),
            UnauthorizedAccessException => (StatusCodes.Status403Forbidden, "The filesystem refused"),
            IOException => (StatusCodes.Status409Conflict, "The filesystem refused"),
            _ => (0, string.Empty),
        };

        // 2. Not one we know: pass it on, so a real bug still shows up as a 500.
        if (status == 0)
        {
            return false;
        }

        // 3. Set the status code on the response.
        httpContext.Response.StatusCode = status;

        // 4. Write the error reply. The reason is the exception's own sentence, and the trace id is the same one the logs carry, so a failure a user
        // reports can be found there.
        return await problems.TryWriteAsync(new ProblemDetailsContext
        {
            HttpContext = httpContext,
            ProblemDetails = new ProblemDetails
            {
                Status = status,
                Title = title,
                Detail = exception.Message,
                Extensions = { ["traceId"] = Activity.Current?.Id ?? httpContext.TraceIdentifier },
            },
        });
    }
}
```

7. A search that stops

The search walks the folders one entry at a time and stops as soon as it has enough matches, and the reply says when it stopped early. Why it matters: a search over a huge drive still answers fast without tying up the server, and the user is told the list was cut short. Docs tab record: Performance, measured. [TheYard: The search index](#)

samples/maplarge/Application/FileBrowser.cs, Search

```
/// <summary>
/// Finds every file and folder under a folder, at any depth, whose name matches the query, and stops once it has the maximum number of matches. The
/// store hands back entries one at a time as it walks, so a search that fills up early never reads the rest of the tree. That early stop and the cap
/// on results keep a search on a large tree fast (more in docs/ADR-008-performance-measured.md).
/// </summary>
/// <param name="path">The folder to search under, relative to home; "" means home.</param>
/// <param name="query">A plain substring, or a glob using * or ?; empty is refused.</param>
/// <param name="limit">
/// The most matches to return. Null uses the configured default, and any value is clamped between 1 and the configured ceiling. </param>
public SearchResult Search(string? path, string? query, int? limit)
{
    long started = Stopwatch.GetTimestamp();
    var pattern = new NamePattern(query);
    if (pattern.IsEmpty)
    {
        throw ApiRefusalException.Refused("A search needs something to look for.");
    }
    int cap = Math.Clamp(limit ?? options.SearchLimit, 1, options.SearchCeiling);
    string absolute = Folder(path);
    var folders = new List<FolderEntry>();
    var files = new List<FileEntry>();
    bool truncated = false;
    foreach (StoreEntry entry in store.Descendants(absolute))
    {
        if (!pattern.Matches(Path.GetFileName(entry.Absolute)))
        {
            continue;
        }
        if (folders.Count + files.Count >= cap)
        {
            truncated = true;
            break;
        }
        Sort(entry, folders, files);
    }
    folders.Sort((a, b) => ViewTotals.NameOrder.Compare(a.Path, b.Path));
    files.Sort((a, b) => ViewTotals.NameOrder.Compare(a.Path, b.Path));
    return new SearchResult(pattern.Query, home.Relative(absolute), folders, files, ViewTotals.Of(folders, files), truncated, Elapsed(started));
}
```

8. main.ts, where the page starts

The page's TypeScript starts here. index.html loads one script, js/main.js, which tsc compiles from src/main.ts. Like Program.cs, it is a table of contents: each line starts one part of the page, in order, with the file that holds the details beside it. Why it matters: a developer sees what the page is made of in one screen and goes straight to the file for the part they need. Docs tab records: One project, four folders, dependencies inward; TypeScript, organised. [TheYard: Projects](#)

samples/maplarge/src/main.ts

```
/**
 * The entry point for the page. It reads like a table of contents: each line starts one part, in order, and the comment beside it names the file
 * that holds the details. Program.cs does the same for the server.
 */

import { currentState, navigate, render, startNavigation } from './navigation.js';
import { connectControls } from './ui/controls.js';
import { findPageElements } from './ui/pageElements.js';
import { showVersion } from './ui/versionFooter.js';

const page = findPageElements(); // ui/pageElements.ts: the elements index.html must have
startNavigation(page); // navigation.ts: the address bar, the two views, Back and Forward
connectControls(page, navigate, currentState); // ui/controls.ts: open, close, the tabs, view links, Escape
showVersion(page.footer, navigate); // ui/versionFooter.ts: the build version in the footer
render(); // navigation.ts: draw what the address asks for
```

9. The address is the state

One function reads the address bar and one writes it. A value that makes no sense falls back to the default, and defaults are left out, so links stay short. Why it matters: an old or mistyped link still opens something sensible instead of an error, and a shared link stays readable. Docs tab record: State lives in the URL. [TheYard: A record with no address](#)

samples/maplarge/src/lib/urlState.ts

```
/**
 * Reads a query string into a State. Unknown keys are ignored. A value that is not in its allowed list falls back to the default. A path is cleaned
 * to forward slashes with no leading or trailing slash, so two ways of writing the same folder give the same State.
 * @param search the query string from the address, with or without the leading "?"
 */
export function parse(search: string): State {
  const params = new URLSearchParams(search.startsWith('?') ? search.slice(1) : search);
  return {
    // Any known key opens the dialog, so a link as short as "?path=docs" works. An open dialog with every other value at its default is written as
    // "?view=browse" by serialize().
    open: KEYS.some((key) => params.has(key)),
    view: pick(params.get('view'), VIEWS, DEFAULTS.view),
    path: cleanPath(params.get('path')),
    q: (params.get('q') ?? '').trim(),
    sort: pick(params.get('sort'), SORTS, DEFAULTS.sort),
    dir: pick(params.get('dir'), DIRS, DEFAULTS.dir),
    doc: (params.get('doc') ?? '').trim().toLowerCase(),
  };
}

/**
 * Writes a State as a query string. Values equal to their default are left out, so links stay as short as possible. A closed State gives "" (no
 * query at all). An open State where every value is the default would also give nothing, so it writes "?view=browse" to keep the dialog open when
 * the address is read back. @returns "" when closed, otherwise "?key=value&..."
 */
export function serialize(state: Partial<State>): string {
  const full: State = { ...DEFAULTS, open: true, ...state, path: cleanPath(state.path) };
  if (!full.open) {
    return '';
  }
  const params = new URLSearchParams();
  for (const key of KEYS) {
    if (full[key] !== DEFAULTS[key] && full[key] !== '') {
      params.set(key, full[key]);
    }
  }
  if (![...params.keys()].length) {
    params.set('view', full.view);
  }
  return `?${params.toString}`;
}
```

10. The one way the state changes

Every change on screen goes through this one function: update the state, write the address, redraw. Closing the dialog is navigating back to the defaults, so Back reopens it where it was. Why it matters: one path for every change means one place to look for a bug, and the Back button always does what a user expects. Docs tab record: State lives in the URL. [TheYard: UI architecture](#)

samples/maplarge/src/navigation.ts, navigate

```
/**
 * Changes the page state. It merges the changed keys into the current state, writes the result into the address bar as a history entry, then
 * redraws. Every state change in the app comes through here. Closing the dialog is also a navigation (to the default state), so pressing Back after
 * closing reopens the dialog where it was.
 * @param partial the state keys that change; keys left out keep their current value
 * @param replaceEntry true to overwrite the current history entry instead of adding one
 * (used while typing in the search box, so each keystroke is not its own Back step)
 */
export function navigate(partial: Partial<State>, replaceEntry = false): void {
  const next: State = { ...state, open: true, ...partial };
  // A closed state serializes to an empty string. pushState with "" would keep the old query string, so the page path with no query string is written
  // instead, which clears it.
  const url = serialize(next) || window.location.pathname;
  if (replaceEntry) {
    window.history.replaceState(null, '', url);
  } else {
    window.history.pushState(null, '', url);
  }
  // Parse the address again instead of keeping `next`. The address is the only source of truth, and parsing it also cleans up any value the
  // serializer left out or would have rejected.
  state = parse(window.location.search);
  render();
}
```